AEROASSISTED MANEUVERS: POTENTIAL AIR AND SPACE LAW CHALLENGES

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ABSTRACT

Diverging from the current paradigm that spacecraft exclusively operate within the vacuum of outer space, current engineering efforts are seeking to create spacecraft that can exploit the aerodynamic forces of the upper atmosphere in order to implement a trans-atmospheric, aeroassisted maneuver. By transcending both the airspace and outer space environments, aeroassisted maneuver represent a source of potential air and space law challenges arising due to current ambiguities surrounding atmospheric delimitation debate as well as the question of airspace sovereignty limits vis-à-vis space law.

I. INTRODUCTION

Spacecraft can be divided functionally into two categories: (1) Vehicles that operate exclusively in the vacuum environment of space; and (2) vehicles that are hybrid in nature and capable of re-entering the Earth's atmosphere following the completion of a given on-orbit mission. While the former category is referred to as

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satellites, the latter are identified trans-atmospheric vehicles, or TAVs. Since the 1970s, TAVs have been limited to boost-glider designs, such as the Space Shuttle and X-37B Orbital Test Vehicle (OTV), which achieve orbit as either a secondary or tertiary stage on a rocket-propelled spacelift system. Following re-entry, the TAV then utilizes aerodynamic lift to perform a gliding maneuver in order to land. Seeking to evolve the boost-glider design concept, the early 21st century has witnessed an emergence of both national and corporate efforts to create a hypersonic spaceplane capable of taking off and landing horizontally on a conventional runway.[1] Designed as a single-stage-to-orbit vehicle, the spaceplane is able to attain a level of mission "availability and flexibility of use" hitherto limited by the mass budget, launch window, and launch site location restrictions inherent in existing rocket booster systems.[2]

Whether designed as a boost-glider or spaceplane, TAVs offer the capability of utilizing the upper atmosphere as an alternative maneuver environment rather than an interface solely for the purpose of re-entry at the mission end-of-life for manned and unmanned spacecraft. Traditionally, orbital states and orbit geometry are modified via various maneuvers performed *in vacuo* which, depending on both the initial mission altitude and desired orbital change, have the propensity of becoming prohibitively expensive in terms of propellant expenditure. Alternatively, atmospheric re-entry can be employed as a means of operational maneuver whereby the aerodynamic drag of the upper atmosphere is exploited by a TAV to create a trans-atmospheric, aeroassisted maneuver. Such maneuvers have been analytically demonstrated to achieve a desired orbital change for less propellant than required by a vacuum-only maneuver, thus extending the spacecraft mission life.

II. TYPES OF AEROASSISTED MANEUVERS

Overall, trans-atmospheric maneuvers can be implemented in three fundamental mission contexts: (1) Orbit modification; (2)

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[1] Carl Q. Christol, *The Aerospace Plane: Its Legal and Political Future*, 9 SPACE POL'Y 35, 36 (1993).

[2] Pierre Betin, *Reflections on the Spaceplane*, 7 SPACE POL'Y 137, 138 (1991).

orbital transfer from High Earth Orbit (HEO) to Low Earth Orbit (LEO); and (3) interplanetary exploration. The first mission context occurs primarily in LEO and is comprised of three distinct maneuvers options, each of which are synergetic in execution since they utilize both atmospheric forces, in the form of aerodynamic drag and lift, and propulsive forces. The first type, known as 'aerobang' maneuvers, consists of a trans-atmospheric flight trajectory augmented by continuous thrusting at maximum throttle so as to limit the duration of atmospheric flight, thereby reducing heat effects produced during re-entry. Similarly, the second type of maneuver, known as 'aerocruise,' also utilizes a steady-state propulsive force during the trans-atmospheric trajectory, but at a throttle level sufficient to counteract only aerodynamic drag. The third maneuver type, known as 'aeroglide,' relies primarily on aerodynamic lift and produces a gliding, unpowered trajectory which employs propulsive forces to only enter and exit the trans-atmospheric trajectory.[3]

The second mission context utilizes an 'aerobrake' maneuver whereby the size and eccentricity of an initial orbit are reduced as a result of aerodynamic drag effects engendered by either a single or successive perigee passages through the upper atmosphere. In addition to aerobraking, the third mission context can also utilize maneuvers called 'aerocapture' and 'aerogravity-assist.' For all three maneuver options, an interplanetary trajectory is designed to intersect the upper atmosphere of a target planet in order to reduce spacecraft energy and establish a stable planetary orbit by exploiting atmospheric drag.[4] Presented only for completeness, the maneuvers comprising the interplanetary mission context are not applicable to the Earth-based orbits discussed below.

III. APPLICABILITY OF AIR AND SPACE LAW

Not officially defined by international treaty, the demarcation between airspace and outer space has created an extant legal

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[3] Richard E. Johnson, Effects of Thrust Vector Control on the Performance of the Aerobang Orbital Plane Change Maneuver (1993) (unpublished MS Thesis, Naval Postgraduate School) (on file with Dudley Knox Library), at 3-4.

[4] Christopher L. Darby & Anil V. Rao, *Optimal Impulsive LEO to LEO Multiple-Pass Aeroassisted Orbital Transfer for Small Spacecraft* 3 (20th AAS/AIAA Space Flight Mechanics Meeting, Paper Presentation, 2010).

debate concerning where air law ends and space law begins. As codified in Article I of the Convention on International Civil Aviation, Chicago, 1944, air law grants each state "sovereignty and exclusive territorial jurisdiction" over its' respective airspace, only to be infringed upon by prior formal agreement or treaty.[5] By contrast, Article II of the Outer Space Treaty of 1967 declares outer space to be an international zone outside the realm of state sovereignty.[6] In order to define the transition between air and space law, two differing methods of airspace and outer space delimitation have arisen: spatialism and functionalism. For spatialists, the boundary between air and space is defined physically in terms of altitude, such as the von Kármán Line devised in the 1950s. Rather than an altitude boundary, the functionalist approach seeks to delimit airspace and outer space according to function and "distinctive traits" of the vehicle operating within the environment in question.[7]

Compounding the demarcation debate is the absence of any conventional or customary rule of international law addressing the "innocent passage" of vehicles ascending to or descending from space. While such passages do occur within foreign airspace, "no protests against [them] have been raised so far" and the passages are viewed as a *fait accompli*.[8] When considered within the context of both the demarcation and atmospheric passage debates, aeroassisted maneuvers pose a series of challenges to air and space law alike. Since aeroassisted maneuvers are initiated from and terminate in space, are they governed by air and/or space law? Can such maneuvers be considered an "innocent passage" when implemented within foreign airspace?

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[5] Alexandra Harris & Ray Harris, *The Need for Air Space and Outer Space Demarcation*, 22 SPACE POL'Y 3, 4 (2006); Stephan Hobe et al., *Space Tourism Activities – Emerging Challenges to Air and Space Law?*, 33 J. SPACE L. 359, 361 (2007).

[6] Harris, *supra* note 5, at 4.

[7] *Id.* at 363.

[8] Andrei D. Terekhov, *Passage of Space Objects through Foreign Airspace: International Custom?*, 25 J. OF SPACE L. 1, at 6, 8 (1997).

A. Spatialism and Aeroassisted Maneuver Altitude Delimitation

Whether an aeroassisted maneuver is implemented to modify an existing orbit or conduct an orbital transfer from HEO to LEO, a trajectory must transit the upper atmosphere at an altitude with sufficient density to impart the requisite aerodynamic force on a TAV. Depending on the desired final orbit geometry as well as the imposition of deceleration and heat transfer constraints by the structure and/or payload of sample TAVs, trajectory simulations have indicated that aeroassisted maneuvers have the potential of occurring within the 50-100 kilometer (km) altitude regime.[9]

When viewed within the context of spatialism, the potential altitude regime for aeroassisted maneuvers occurs lower than many international attempts to delimit airspace and outer space. Corresponding to an altitude of 100 km, the von Kármán Line represents an approximate boundary above which an aircraft cannot derive any aerodynamic lift from the atmosphere and must travel at a speed approaching orbital velocity.[10] As an alternative, Italy in 1975 proposed a boundary at 90 km since it represented the median altitude between the upper limit of aircraft flight at 60 km, and the lowest possible satellite orbit at 120 km. In 1976, Belgium echoed the reasoning of von Kármán by advocating a boundary at 100 km, while the Soviet Union in 1979 proposed an arbitrary boundary "at an altitude not exceeding 110 km."[11]

Although the Italian delimitation proposal places the upper altitude limit of aircraft flight at 60 km, this corresponds to the approximate operating altitude of the X-15, an experimental rocket-propelled aircraft of the early 1960s.[12] In terms of conventional

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[9] Christopher L. Darby & Anil V. Rao, *Minimum-Fuel Low-Earth Orbit Aeroassisted Orbital Transfer of Small Spacecraft*, 48 J. SPACECRAFT & ROCKETS 618-628 (2011); Patrick R. Jolley & Stephen A. Whitmore, *Aerodynamic and Propulsion Assisted Maneuvering for Orbital Transfer Vehicles* (5th Responsive Space Conference, Paper Presentation, 2007), at 1-39.

[10] FRANCIS LYALL & PAUL B. LARSEN, SPACE LAW: A TREATISE 167-168 (Ashgate Publishing Limited 2009).

[11] *Id.*, at 169.

[12] W. D. Kay, *The X-15 Hypersonic Flight Research Program: Politics and Permutations at NASA*, *in* FROM ENGINEERING SCIENCE TO BIG SCIENCE: THE NACA AND NASA COLLIER TROPHY RESEARCH PROJECT WINNERS 155 (Pamela E. Mack, ed., U.S. GPO 1998).

aircraft, the upper altitude limit is considerably lower with the U.S. Air Force's U-2 reaching a maximum ceiling of approximately 21 km. As for spacecraft, the lowest operational orbit corresponds to an altitude of 96 km, which is lower than all aforementioned delimitation proposals.[13] By considering both the nominal ceiling of the X-15 and the lowest achieved satellite orbit, an altitude "gap" at 60-96 km is created which aligns with the potential operating environment for aeroassisted maneuvers.

Even though conventional aircraft and lighter-than-air vehicles such as blimps do not operate at 60-96 km, this altitude regime is still considered to be sovereign airspace as evidenced in several reports issued since the 1960s. As one example, the Canadian government identified that the Space Shuttle *Challenger* flew within its airspace at an altitude of approximately 68 km while on glide path to land in the United States following re-entry in 1984.[14] A second example arises from the proceedings of the United Nations' Committee on the Peaceful Uses of Outer Space in 1996.[15] In response to a questionnaire disseminated to member states regarding legal issues associated with "aerospace objects," Germany noted that the Soviet Space Shuttle *Buran* passed through the airspace of Turkey following re-entry in 1988.[16] Based on a similar structural design and mission profile, the *Buran* is assumed to have flown through Turkish airspace at an altitude commensurate with that of the *Challenger*.

Despite the ambiguity surrounding the actual spatial delimitation of airspace and outer space, precedence dictates that airspace sovereignty extends up to and beyond an altitude of 90 km. Consequently, aeroassisted maneuvers occurring at an altitude 50-90 km would be considered a passage through foreign airspace if not implemented over international waters.[17] By implementing

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[13] Katherine M. Gorove, *Delimitation of Outer Space and the Aerospace Object – Where is the Law?*, 28 J. SPACE L. 11, 12 (2000).

[14] Terekhov, *supra* note 8.

[15] For the complete questionnaire, *see* Gorove, *supra* note 13, at 17-18; for the complete member states' responses to the questionnaire, *see also* U.N. Committee on the Peaceful Uses of Outer Space, *Questionnaire on Possible Legal Issues: Replies from Member States*, UN Doc. A/AC.105/635, (Feb. 15, 1996).

[16] UN Doc. A/AC.105/635, *supra* note 15, at 7.

[17] Since aeroassisted maneuvers implemented over international represent benign occurrences, only those maneuvers that infringe on foreign airspace will be considered henceforth.

an aeroassisted maneuver within airspace, a TAV is then subject to the jurisdiction of air law. Since an aeroassisted maneuver places a TAV within foreign airspace for a time of finite duration however, can the passage and airspace infringement be deemed "innocent" and then overlooked by the overflown state as with cases of vehicles ascending to or descending from space? The presumptive answer would be in the affirmative, but a functional analysis of the TAV mission is required in order properly classify an aeroassisted maneuver as an "innocent passage" or not.

B. Functionalism and TAV Classification

According to the functionalist approach, the question of legal jurisdiction is dependent on the function of the vehicle in question. Outlined in the 1975 Convention on Registration of Objects Launched into Outer Space, a launch vehicle and satellite payload are considered "space objects" and therefore governed by space law since they are intended to reach and operate within the space environment.[18] A broad term, "launch vehicle" within the context of the aforementioned Convention applies to rocket boosters and not carrier aircraft. For the latter case, such as the Lockheed L-1011 transport aircraft utilized as an upper atmospheric launching platform for the Pegasus booster rocket, both the carrier aircraft and attached spacecraft are governed by air law until vehicle separation. Following separation, the Pegasus booster and similar spacecraft cannot "derive support in the atmosphere from the reactions of the air" and are thus considered "space objects" subject to space law.[19]

Based on the Convention on Registration, a TAV conducting normal mission operations in orbit is considered a "space object" and is subject to space law. When conducting an aeroassisted maneuver however, the TAV utilizes aerodynamic forces within the upper atmosphere to produce lift. Does this ability to leverage aerodynamic forces during the aeroassisted maneuver necessitate a change in vehicle status from space object to aircraft, and a like-

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[18] Convention on Registration of Objects Launched into Outer Space, *opened for* signature Jan. 14, 1975, 28 U.S.T. 695, 1023 U.N.T.S. 15.

[19] JANE VAN NIMMEN, ET AL., NASA HISTORICAL DATA BOOK, VOLUME VII: NASA LAUNCH SYSTEMS, SPACE TRANSPORTATION, HUMAN SPACEFLIGHT, AND SPACE SCIENCE, 1989-1998 55 (U.S. GPO 1999); Hobe, *supra* note 5, at 364.

wise change in legal jurisdiction from space law to air law? Since the TAV produces lift while transiting the upper atmosphere, then it could be assumed that air law supersedes any space law consideration as with the preceding example of the Pegasus booster attached to the L-1011 carrier aircraft. The validity of this assumption is tenuous, especially when the functions of both the TAV and aeroassisted maneuver are considered. With the former, a TAV is intended to reach and operate within outer space and thus constitutes the baseline definition of a "space object." For the latter, an aeroassisted maneuver is implemented in order to alter the geometry of an orbit, whether originally in LEO or HEO. As a result, the TAV always remains within the space environment except for the duration of the aeroassisted maneuver itself (and the eventual re-entry at mission end-of-life).

If a TAV is subject to air law during an aeroassisted maneuver, then the right of foreign airspace sovereignty must be observed. Consequently, a state whose airspace will be infringed by an aeroassisted maneuver maintains the right to regulate passage within its airspace. Apart from civilian missions such as those related to science or transportation, as with the case of space tourism, TAVs also have the potential of hosting a variety of military functions. From being a platform for augmented command, control, communications, intelligence, surveillance, and reconnaissance (C3ISR), or a vehicle for prompt global strike, TAVs proffer an undeniable enhancement of military capabilities.[20] Based on the their inherent military mission implications, aeroassisted maneuvers could be implemented to either deliver a TAV over a target of interest, or place a TAV within the atmosphere to conduct a specific mission within the airspace of a state being overflown.

In light of these potential missions, a state could follow precedence and impose a "no-fly zone" for aeroassisted maneuvers deemed to fall outside the bounds of an "innocent passage." For example, the French and Spanish governments imposed "no-fly zones" which prevented the passage of U.S. Air Force aircraft through their respective airspaces when executing Operation El

[20] Jinyuan Su, *Near Space as a* Sui Generis *Zone: A Tri-Layer Approach of Delimitation*, 29 SPACE POL'Y 90, 91 (2013).

Dorado Canyon against Libya in 1986.[21] Similarly, a state could impose a "no-fly zone" precluding an aeroassisted maneuver intended to insert a TAV in orbital position to complete a specific military mission, e.g. C3ISR or prompt global strike. When considered under the jurisdiction of air law, aeroassisted maneuvers implemented in violation of a state-imposed "no-fly zone" would therefore constitute a breach of international treaty.

IV. ENVIRONMENTAL CONSIDERATIONS

Occurring within the 50-90 km altitude regime, aeroassisted maneuvers place a TAV not only within potential foreign airspace, but also in the physical environment of the upper atmosphere. Of the various human space activities, space launch produces a high level of exhaust pollutants in the form of dust, the emission of toxic compounds such as aluminum oxide (from solid propellant), and the spraying of unburned liquid propellant like hydrazine. Although argued by many to have a negligible cumulative effect on atmospheric degradation, the burning of rocket propellant – whether solid or liquid in composition – in the upper atmosphere has been demonstrated to deteriorate the ozone layer and chemically contaminate the water cycle.[22] Not chemically destructive, the release of water as an exhaust by-product can interfere with ionospheric conditions, thus disrupting the transmission of wireless communications.[23]

While the aeroglide subtype of aeroassisted maneuver only performs thruster burns in space, the aerobang and aerocruise alternatives produce a steady thrust throughout the trans-atmospheric trajectory. Even though a TAV's propulsion system burns liquid rather than solid propellants and therefore produces fewer pollutants, exhaust by-products are still be continuously injected into the airspace when an aerobang or aerocruise maneuver is implemented. As a result, can a state deny the infringement of its airspace by an aeroassisted maneuver due to environmental considerations? With the aerobang or aerocruise maneuver occur-

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[21] JOSEPH T. STANIK, EL DORADO CANYON: REAGAN'S UNDECLARED WAR WITH QADDAFI 145-146 (Naval Institute Press 2003).

[22] LOTTA VIIKARI, THE ENVIRONMENTAL ELEMENT IN SPACE LAW: ASSESSING THE PRESENT AND CHARTING THE FUTURE 29-31 (Koninkliijke Brill NV 2008).

[23] *Id.* at 31.

ring within the jurisdiction of air law, does the operator of the TAV assume sole liability for any environmental impact of the maneuver?

V. CONCLUSION

The continued engineering development of TAVs will undoubtedly require the air and space law challenges of aeroassisted maneuvers to be formally addressed due to ongoing debate associated with the prospect of airspace and outer space delimitation. Occurring within the 50-90 km altitude regime, spatialism dictates that TAVs implementing an aeroassisted maneuver are subject to air law. From the functionalist perspective however, the legal delimitation of air and space law becomes ambiguous with arguments that can identify a TAV as either an aircraft or space object. For many states, to include the Lebanon, the Syrian Arab Republic, and Turkey, the stance is clear: A TAV traversing foreign airspace during an aeroassisted maneuver is subject to air law. For states like the Czech Republic however, ambiguity resurfaces with the view that air law only applies to "objects resembling [spaceplanes], but not to objects resembling Space Shuttles."[24]

Due to the unique hybrid characteristics of not only aeroassisted maneuvers, but also TAVs, one viable solution option is to spatially establish an exclusive zone of operation for TAVs between the maximum operating ceiling of conventional aircraft and the lowest achievable orbit for satellites. Defined as a "*sui generis* zone," the approximate altitude regime of 21-96 km would permit the freedom of operation of TAVs (within peaceful bounds) and officially delimit the boundaries of both airspace sovereignty and outer space.[25] A propitious compromise, such an exclusive zone for TAVs could forestall the onset of legal challenges for nascent commercial and national ventures seeking to implement aeroassisted maneuvers.

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[24] Gorove, *supra* note 13, at 21-22.
[25] Su, *supra* note 20, at 92.